

# Tinglin Garden: Map & Memory

## 亭林园: 地图与记忆

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### Abstract

#### Between Map and Memory: An Interactive Map and VR-Based Exploration of Tinglin Garden

Our project aims to design an interactive map and VR tour guide system for Kunshan Tinglin Garden, addressing the issue of its official website lacking overall spatial display and virtual tour functions. The system is divided into the “map cognition layer” (map) and the “immersive experience layer” (memory). As a bridge between Kunshan’s local heritage and our international community at DKU, our goal is to lower the cultural threshold of this historic landmark. We aim to help both local and non-local visitors, especially international visitors who face language and spatial barriers, to navigate the physical park while accessing its deep cultural narratives. By integrating spatial data, historical texts, and on-site collected images, the system will enhance tourists’ spatial cognition and cultural understanding, and also promote public participation.

#### Disclaimer

This InfoVis project is created for submission to INFOSCI 301: Data Visualization and Information Aesthetics (Spring 2026), instructed by Professor Luyao Zhang. The project is solely for academic purposes. The DKU Community-based learning program supports this course.

#### Acknowledgments

We thank Professor Luyao Zhang for her guidance, our INFOSCI 301 classmates for iterative feedback, and the Jiading Urban Exhibition Hall and Kunshan Museum of Chinese Opera Genres for providing valuable field insights. We also appreciate the industry and community contributors who shared perspectives on the interactive map and local urban development.

#### Statement of Accessibility

The project repository and documentation are available at:

GitHub: <https://github.com/PandaSun-43/INFOSCI301-Final-Project-Keviwen>

We follow accessibility guidelines recommended by SIGCHI: <https://sigchi.org/resources/guides-for-authors/accessibility/>

#### Keywords

Information Visualization; Geospatial Storytelling; Community-Based Learning; Digital Cultural Heritage; Interactive Maps; Immersive VR; UN SDGs.

### Contribution to SDG Goals

#### SDG 4 - Quality Education

- Lowering cultural thresholds for international visitors.

#### SDG 9 - Industry, Innovation, and Infrastructure

- Using open-source tools for low-cost digital innovation.

#### SDG 11 - Sustainable Cities & Communities

- Digital preservation of urban green spaces & heritage.

#### SDG 17 - Partnerships for the Goals

- Connecting campus resources with local communities.



### Data Governance Principles

#### Transparency

- Clear data sources and citations
- Adheres to FAIR principles: Findable, Accessible, Interoperable, Reusable

#### Privacy

- No identifiable personal data
- Adheres to CARE’s Responsibility and Ethics

#### Inclusivity

- Supports multiple languages and perspective
- Contextualizes local knowledge for non-local users

#### Accessibility

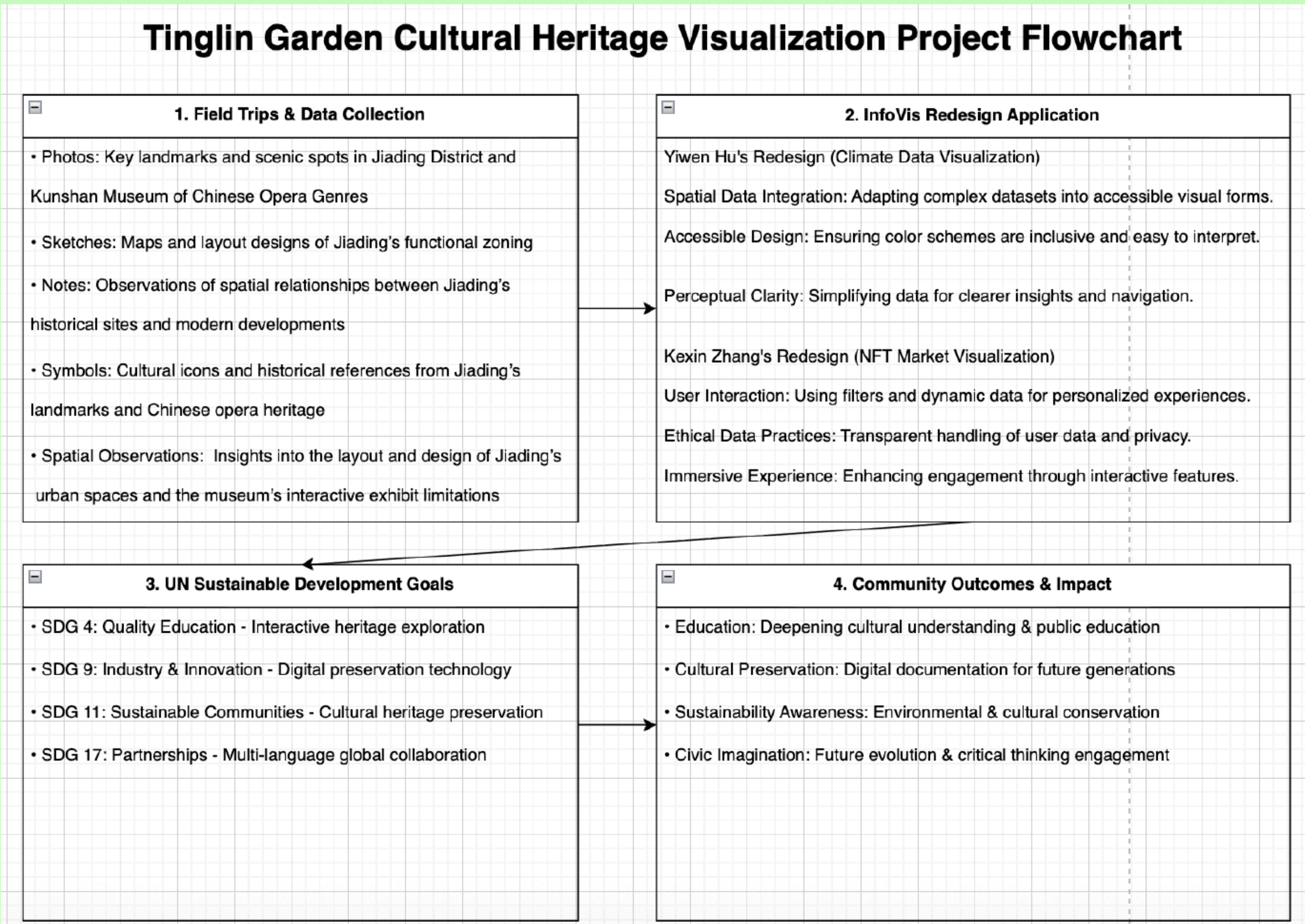
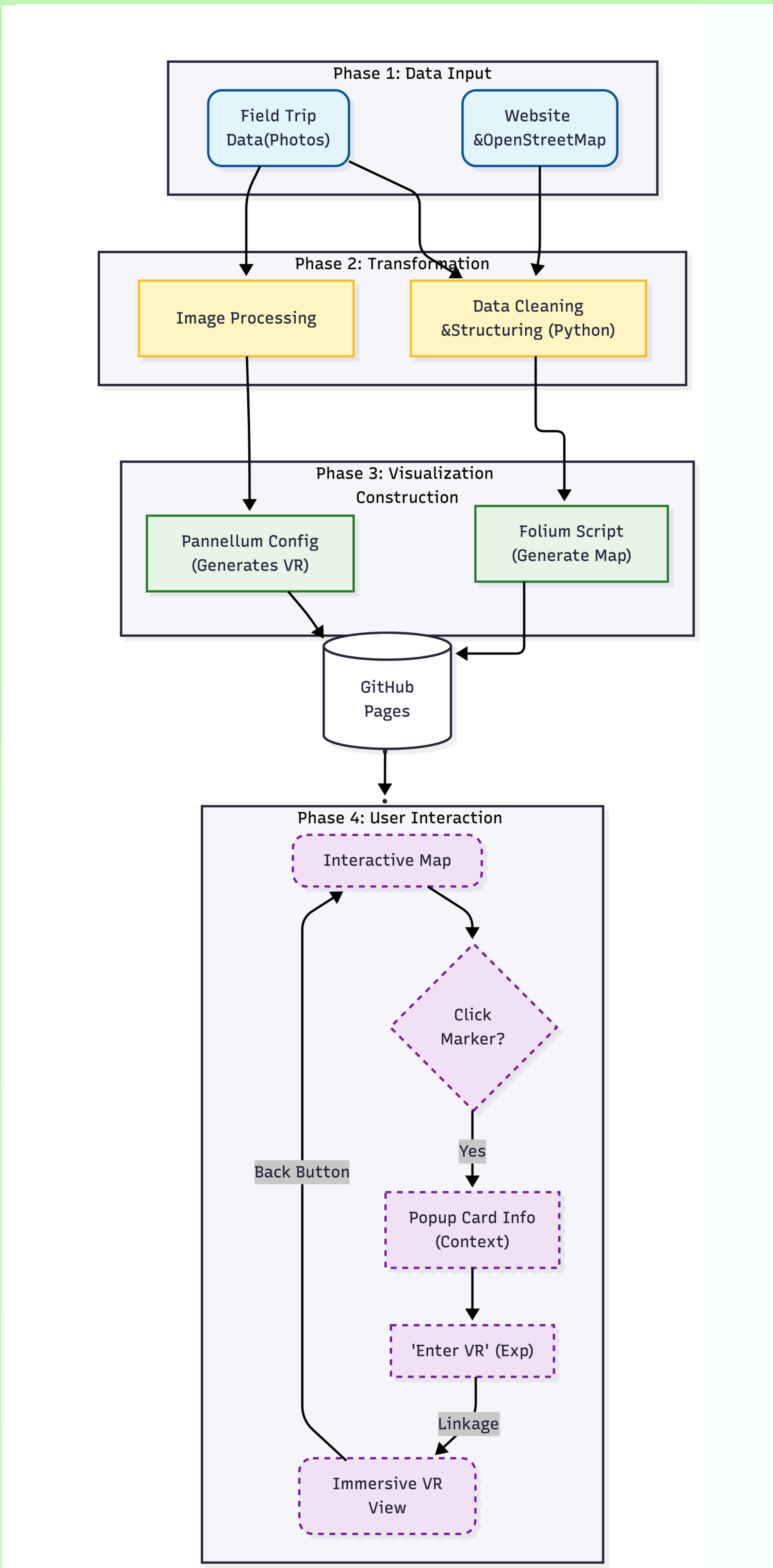
- Data in open formats (CSV/JSON/JPG)
- Public GitHub documentation

### System Architecture Design

Our system adopts a hybrid, client-side architecture designed to support seamless geospatial navigation and immersive cultural storytelling. The frontend is built with HTML and Python, integrating interactive elements and smooth routing into the geospatial map. For map rendering, we use Folium, which interfaces with OpenStreetMap for geospatial tiles, ensuring spatial accuracy and flexibility.

The backend consists of a Python-based data processing pipeline that handles geospatial data, historical narratives, and field-collected media (photos, 360° panoramas). The system includes custom HTML-based pop-up cards for each site, displaying detailed cultural information and images.

We use external geospatial data from OpenStreetMap and Tinglin Garden’s historical archive, which is preprocessed and directly embedded into the map. The project is hosted on GitHub Pages, with the static content ensuring a lightweight, accessible, and portable cultural mapping prototype. Additionally, bilingual labeling is incorporated to cater to both local and international users, promoting a more inclusive experience.



### App Flow Diagram with Functional Decomposition

#### System Workflow:

- Data Preparation: We collect geospatial data from OpenStreetMap, historical texts from Tinglin Garden's official archive, and photos from fieldwork.
- Map Rendering: The core interactive map is built using Folium, with custom markers and pop-up cards to display site-specific cultural and historical information.
- Pop-up Cards: These cards are composed of HTML and CSS, incorporating base64-encoded images for compatibility and offline use. Each card includes rich cultural context and imagery, providing an engaging storytelling experience.
- Data Preprocessing: The system processes GPS coordinates, metadata, and image files to structure inputs for the interactive map. We also use Python scripts to validate data and standardize the content, ensuring the map is accurate and functional.
- User Interaction: Users can navigate the map by panning and zooming. Hovering over markers displays quick previews, while clicking a marker brings up a full cultural story card with photos and text. The map includes filtering options for a more focused exploration (e.g., filtering by historical sites or cultural significance).

#### Technical Pipeline:

The design process includes using Plotly for prototyping and testing spatial layouts and color schemes. After refining the design, it is transferred into Folium for final rendering. The interaction flow begins with the map, allowing users to explore spatial data, and transitions into detailed cultural narratives via pop-up cards.

#### Final Visualization:

The project is hosted on GitHub Pages and deployed as a standalone HTML file. This ensures the system is fully accessible across various devices and browsers, aligning with the course’s emphasis on creating interactive and community-driven visualizations.



### Team Makeup & Strengths

#### Roles & Responsibilities

##### Coding & Engineering -- Yiwen Hu

- Handles coding, geospatial integration, and visualization design. Develops the interactive map, VR experiences, and user interface, ensuring smooth navigation and immersion.

##### Data Collecting & Interaction Design -- Kexin Zhang

- Collects, formats, and curates historical texts, field-collected images, and geospatial data. Ensures data quality and prepares it for integration into the map and VR system.

##### Shared Roles

- Interaction Testing & Iteration
- Documentation Lead (co-maintaining GitHub, writing clear user-facing descriptions)
- Community Liaison (linking design choices to community-based learning values)

##### Complementary Skills

Our collaboration merges data handling, geospatial reasoning, visual storytelling, and UI/UX design. Techniques learned in the Infovis Redesign Project, such as multivariable visual encoding, narrative structuring, and simplifying complex information, directly inform our restaurant cards, cultural-story annotations, and thematic routes. Together, we create an interactive map that is technically robust, culturally meaningful, and ethically grounded.



Scan here to check the GitHub code of our project.